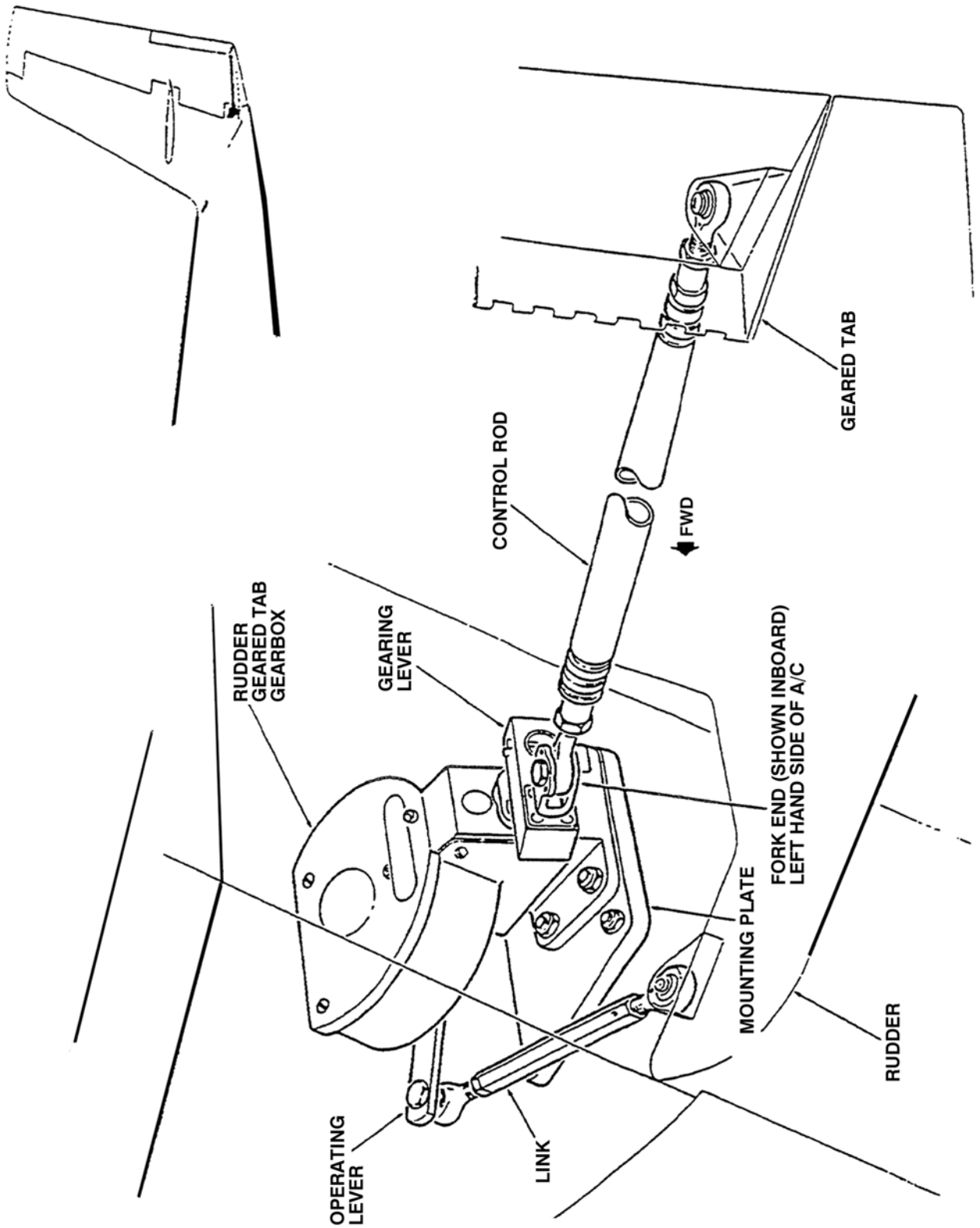
Rudder Geared Tab System – General Data

1. General

A geared tab is installed on the lower portion of the rudder trailing edge to provide aerodynamic assistance to rudder movement. The geared tab system consists of control rods, levers, gearbox and tab. The gearbox, which provides a non-linear deflection of the tab in relation to rudder movement, is bolted to a mounting plate installed on the bottom rudder hinge bracket of the vertical stabilizer. A link connects the rudder to the gearbox operating lever, and the gearbox gearing lever is connected to the tab by a control rod. Movement of the rudder is transmitted through the gearbox, where the mechanism changes the ratio of movement of tab to rudder, as it moves the tab in the opposite direction to the rudder. Refer to Figure 1 for details.



Rudder Geared Tab System
Figure 1

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2. Adjustment/Test

A. Rig Rudder Geared Tab

Refer to Figure 1 for details.

1) Check that geared tab is neutral with rudder neutral (refer to 27-20-00, Rudder Control System – General Data).

2) Check that gearbox gearing lever is horizontal with fork end inboard, and that rod attachment bolt is vertical.

3) Adjust link to obtain gearing lever position, then adjust control rod to set tab in neutral.

4) Tighten and wirelock control rod lock nuts as necessary.

5) Check rudder geared tab control surface clearances (refer to 27-00-00, Flight Controls – General Data).

6) Operate rudder and check for full and free range of movement (refer to 27-20-00, Rudder Control System – General Data). Make sure that the direction of geared tab movement is opposite rudder movement.

3. Inspection/Check

A. Free Play Check – Rudder Geared Tab Gearbox

Refer to Figure 1 for details.

1) Remove rudder gearbox access panels at rear fuselage.

2) Disconnect control rod from fork end, and operating lever from link.

3) Move gearing lever fore and aft and check that free play at shaft of gearing lever does not exceed 0.002 inch. Move operating lever up and down and check that free play at gearbox bolt does not exceed 0.002 inch.

4) Reconnect operating lever to link, and control rod to fork end.

5) Install rudder gearbox access panels.

B. Free Play Check – Rudder Geared Tab

1) Operate rudder (and geared tab) to neutral position.

2) Move geared tab manually and check that free play (backlash), measured at trailing edge, does not exceed 0.10–inch.

Note Measurement to be taken adjacent to control rod attachment point.

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